Analytic Calculation of Volumes

General formulation and change to Cylindrical Coordinates

For $k, R, r > 0$, the sphere and torus in Cartesian coordinates are

$$S = \left\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq k^2\right\}$$
$$T = \left\{(x, y, z) \in \mathbb{R}^3 : (\sqrt{x^2 + y^2} - R)^2 + z^2 \leq r^2\right\}$$

The desired volume is $V = \iiint_{S \cap T} 1 \, dx \, dy \, dz$. We introduce cylindrical coordinates

$$\Phi(\rho, \theta, z) = (\rho \cos\theta, \rho \sin\theta, z)$$

On $A = (0, \infty) \times (0, 2\pi) \times \mathbb{R}$, this is a C^1-diffeomorphism onto $\mathbb{R}^3 \setminus H$, where $H = \{(x, 0, z) : x \geq 0\}$. Its Jacobian is

$$D\Phi(\rho, \theta, z) = \begin{pmatrix} \cos\theta & -\rho\sin\theta & 0 \\ \sin\theta & \rho\cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad |\det D\Phi(\rho, \theta, z)| = \rho$$

The shapes in these coordinates are

$$S' = \left\{(\rho, \theta, z) \in A : \rho^2 + z^2 \leq k^2\right\}$$
$$U' = \left\{(\rho, \theta, z) \in A : (\rho - R)^2 + z^2 \leq r^2\right\}$$

The change-of-variables formula gives

$$\boxed{V = \iiint_{S \cap T} 1 \, dx \, dy \, dz = \iiint_{S' \cap U'} \rho \, d\rho \, d\theta \, dz}$$

Case 1b: $k = 1$, $R = 0.5$, $r = 0.5$

In this case we have $T \subseteq S$: Let $(x, y, z) \in T$ and $\rho = \sqrt{x^2 + y^2}$. By the defining inequality for T,

$$\sqrt{(\rho - 0.5)^2 + z^2} \leq 0.5$$

The triangle inequality in $\mathbb{R}^2$ gives

$$\begin{aligned} \sqrt{x^2 + y^2 + z^2} = \|(\rho, z)\|_2 &= \|(\rho - 0.5, z) + (0.5, 0)\|_2 \\ &\leq \sqrt{(\rho - 0.5)^2 + z^2} + 0.5 \leq 1 \end{aligned}$$

So $(x, y, z) \in S$, proving $T \subseteq S$ and $S \cap T = T$. As a result we can use the formula of the volume of a Torus $\mathrm{Vol}(T) = 2\pi^2 R r^2$,

$$\boxed{V = 2\pi^2(0.5)(0.5)^2 = \frac{\pi^2}{4} \approx 2.4674011003}$$

Case 1a: $k = 1$, $R = 0.75$, $r = 0.4$

We will compute the volume by selecting integration boundaries and then applying Fubini's theorem to reduce the triple integral to one-dimensional integrals.

We notice that in the inequalities that define the shapes there is no θ. That is because both solids are symmetric around the z-axis. We can therefore use the following bound $0 \leq \theta \leq 2\pi$ and determine the bounds of the remaining variables in the (ρ, z)-plane.

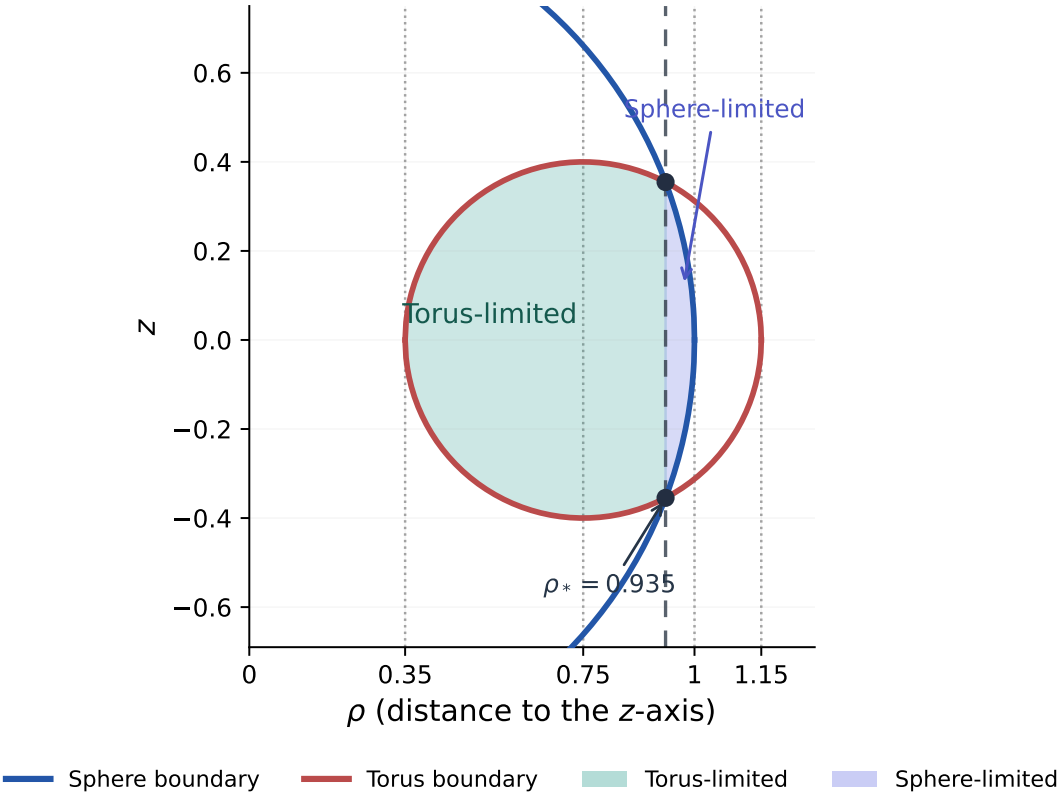


Figure 1: Case 1a. Cross-section in the (ρ, z)-plane. The torus occupies $\rho \in [0.35, 1.15]$, the sphere occupies $\rho \in [0, 1]$, and so the overlap satisfies $\rho \in [0.35, 1]$. For $\rho \in [0.35, 0.935]$ the intersection is bounded vertically by the torus, while for $\rho \in [0.935, 1]$ it is bounded vertically by the sphere.

As far as ρ is concerned it is obvious in Figure 1 that $0 \leq \rho \leq 1$ for the sphere and $0.35 \leq \rho \leq 1.15$ for the torus, and so the bounds for the intersection are $0.35 \leq \rho \leq 1$.

For fixed ρ, the vertical bounds are obtained by solving each inequality for z:

$$-\sqrt{1-\rho^2} \leq z \leq \sqrt{1-\rho^2}$$

for the sphere, and

$$-\sqrt{0.16-(\rho-0.75)^2} \leq z \leq \sqrt{0.16-(\rho-0.75)^2}$$

for the torus.

So, for each fixed ρ, the upper bound is the smaller of

$$h_s(\rho) = \sqrt{1-\rho^2}, \qquad h_t(\rho) = \sqrt{0.16-(\rho-0.75)^2}$$

The transition occurs where $h_s(\rho) = h_t(\rho)$, meaning

$$1-\rho^2 = 0.16-(\rho-0.75)^2$$

which simplifies to

$$1.4025 - 1.5\rho = 0$$

and

$$\boxed{\rho_* = 0.935}$$

Therefore,

$$h_t(\rho) \leq h_s(\rho) \quad \text{for } 0.35 \leq \rho \leq 0.935,$$

and

$$h_s(\rho) \leq h_t(\rho) \quad \text{for } 0.935 \leq \rho \leq 1$$

So the integral must be split into two parts. By Fubini's theorem,

$$V = \int_0^{2\pi} \left[\int_{0.35}^{0.935} \int_{-\sqrt{0.16-(\rho-0.75)^2}}^{\sqrt{0.16-(\rho-0.75)^2}} \rho \, dz \, d\rho + \int_{0.935}^{1} \int_{-\sqrt{1-\rho^2}}^{\sqrt{1-\rho^2}} \rho \, dz \, d\rho \right] d\theta$$
$$= 4\pi \left[\int_{0.35}^{0.935} \rho\sqrt{0.16-(\rho-0.75)^2} \, d\rho + \int_{0.935}^{1} \rho\sqrt{1-\rho^2} \, d\rho \right]$$

The remaining integrals are easy to calculate so finally we have

$$V = 4\pi \left[\int_{0.35}^{0.935} \rho\sqrt{0.16-(\rho-0.75)^2} \, d\rho + \int_{0.935}^{1} \rho\sqrt{1-\rho^2} \, d\rho \right] \approx 1.8560558870$$

$$\boxed{V \approx 1.856056}$$

Case 3: Off-center torus, $k = 1$, $R = 0.75$, $r = 0.4$

The torus is shifted by 0.1 along the z-axis. In Cartesian and cylindrical coordinates, respectively, its regions are

$$T_c = \left\{(x, y, z) \in \mathbb{R}^3 : (\sqrt{x^2+y^2} - 0.75)^2 + (z-0.1)^2 \le 0.16\right\}$$
$$U_c' = \left\{(\rho, \theta, z) \in A : (\rho - 0.75)^2 + (z-0.1)^2 \le 0.16\right\}$$

As in Section 3, $0 \le \theta \le 2\pi$ and $0.35 \le \rho \le 1$. The shift breaks symmetry on $z = 0$, so the upper and lower bounds must be considered separately.

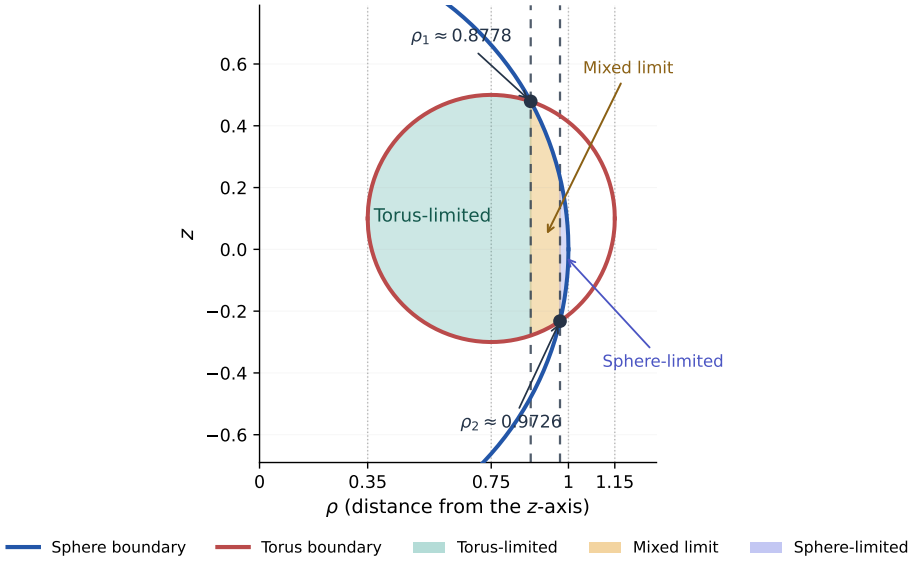


Figure 2: Case 3. Cross-section in the (ρ, z)-plane. The vertical bounds change at $\rho_1 \approx 0.8778$ (upper bound) and $\rho_2 \approx 0.9726$ (lower bound).

We define the sphere and torus half-heights by

$$s(\rho) = \sqrt{1-\rho^2}, \qquad t(\rho) = \sqrt{0.16-(\rho-0.75)^2}$$

The vertical bounds are $[-s, s]$ for the sphere and $[0.1-t,\ 0.1+t]$ for the torus. Their boundary curves intersect where

$$\rho^2 + z^2 = 1 \qquad \text{and} \qquad (\rho-0.75)^2 + (z-0.1)^2 = 0.16$$

Subtracting gives $1.5\rho + 0.2z = 1.4125$. Solving the system gives

$$\rho_1 \approx 0.8777951820 \ (z > 0), \qquad \rho_2 \approx 0.9726414992 \ (z < 0)$$

As shown in Figure 2, both vertical bounds come from the torus on $[0.35, \rho_1]$. The lower bound comes from the torus and the upper from the sphere on $[\rho_1, \rho_2]$. And both come from the sphere on $[\rho_2, 1]$. Therefore, integrating

first with respect to z and θ as in Section 3, Fubini's theorem gives

$$V = 2\pi\left[2\int_{0.35}^{\rho_1} \rho\, t(\rho)\, d\rho + \int_{\rho_1}^{\rho_2} \rho\big(s(\rho) + t(\rho) - 0.1\big)\, d\rho + 2\int_{\rho_2}^{1} \rho\, s(\rho)\, d\rho\right]$$

Using the antiderivatives which are now not hard to find we get

$$\boxed{V \approx 1.8283804582}$$